MIL-R-14368C
5 September 1974
SUPERSEDING
MIL-R-14368B
8 November 1963

MILITARY SPECIFICATION

# REGULATOR ASSEMBLY, RELAY ASSEMBLY, AND VOLTAGE REGULATOR ASSEMBLY: 24-VOLT (NOMINAL RATING), 100- TO 400-AMPERE, DIRECT CURRENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of waterproof, 24-volt (nominal rating), direct current, 100- to 400-ampere, generator regulator assembly intended to regulate the output of generators used on military vehicles. This specification also covers the two principal replaceable component assemblies; the relay assembly and the voltage regulator assembly.

1.2 Classification. Items described in this specification shall be classified in the following manner:

| | | |
|---|---|---|
| Unit 1 | - | Regulator assembly (includes units 2 and 3). |
| Unit 2 | - | Relay assembly. |
| Unit 3 | - | Voltage regulator assembly. |

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.



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SPECIFICATIONS

Federal

| | | |
|---|---|---|
| QQ-P-416 | - | Plating, Cadmium (Electrodeposited). |
| QQ-S-365 | - | Silver Plating, Electrodeposited; General Requirements. |
| QQ-Z-325 | - | Zinc Coating, Electrodeposited, Requirements for. |

Military

| | | |
|---|---|---|
| MIL-W-5086 | - | Wire, Electric-Hookup and Interconnecting, Polyvinyl Chloride-Insulated, Copper or Copper Alloy Conductor. |
| MIL-B-11188 | - | Battery, Storage, Lead-Acid. |
| MIL-G-12604 | - | Generator: Engine Accessory, Direct Current, 28-Volt Nominal. |
| MIL-F-13927 | - | Fungus Resistance Test, Automotive Components. |
| MIL-S-19500 | - | Semiconductor Device, General Specification for. |

STANDARDS

Military

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - | Marking for Shipment and Storage. |
| MIL-STD-130 | - | Identification Marking of U. S. Military Property. |
| MIL-STD-202 | - | Test Methods for Electronic and Electrical Component Parts. |
| MIL-STD-461 | - | Electromagnetic Interference Characteristics Requirements for Equipment. |
| MS-51005 | - | Regulator, Engine Generator: 24-Volts D.C.: 150-400 Ampere, Carbon Pile. |

DRAWINGS

Ordnance Corps

| | | |
|---|---|---|
| D8689197 | - | Cover, Generator Control Box. |
| D8689199 | - | Case, Generator Control Box. |
| D8689245 | - | Relay Assembly. |
| F8723894 | - | Control Assembly, Generator. |
| F8723895 | - | Voltage Regulator Assembly. |
| B8724048 | - | Plate, Name. |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.



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NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government P inting Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 First article. A first article shall be produced and furnished for testing prior to the manufacture of regulator assemblies in production quantities. The First Article(s) shall pass testing specified herein (see 4.2) All items furnished to the Government under this specification shall be representative of those which have passed First Article Testing and inspection. Approval of the first article samples shall not relieve the contractor of his responsibility to provide regulators in production quantities that a·e fully representative of the first article samples. Changes o·· deviations of the production units from the first article sample shall be subject to approval of the contracting officer.

3.2 Materials. Materials shall be as specified herein, on applicable drawings or standards, and in applicable specifications.

3.3 Design and construction.

3.3.1 Use of dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided. Where such contact is not necessary to complete an electrical circuit, but is otherwise unavoidable, parts shall be insulated.

3.3.2 Interchangeability. All units shall be mechanically and electrically interchangeable with all other such units of the same MS Part No. (Unit 1-MS51005 and Drawing F6723894; Unit 2-Drawing D6689245; Unit 3-Drawing F6723895) previously furnished for this application.

3.3.3 Threaded parts. All screw threads shall conform to Handbook H28 (see 2.2). The form, numbe· per inch, and class of threads shall be as specified on applicable standards or drawings.

3.3.4 Connectors. Except as otherwise specified, each unit shall be provided with electrical connecto s of the type and style specified on the applicable drawing.

3.3.5 Cable. Unless otherwise specified (see 6.1), all cable shall conform to Specification MIL-W-5086, type II.

3.3.6 Polarity. All units shall operate in circuits having the battery negative terminal grounded.



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3.3.7 Contact points. The contact points incorporated in the reverse current cutout relay shall not fuse together nor pit as a result of heat or impact incident to their normal operation.

3.4 Performance. The following requirements are applicable to units as specified. When unit 1 is specified it shall be tested to determine conformance to 3.4.1 and 3.4.2. When unit 2 or 3 is specified it shall be installed as a component of unit 1 and tested to determine conformance to 3.4.1 or 3.4.2, as applicable. The unit 1 thus employed shall have previously met the full requirements of this specification.

3.4.1 Relay assembly performance. (units 1 and 2 only).

3.4.1.1 Closing voltage. When tested as specified in 4.6.1.1.1, the relay assembly shall connect the generator to the battery on rising generator output voltage within the areas indicated in figure 1.

3.4.1.2 Opening current. When tested as specified in 4.6.1.1.2, the relay assembly shall disconnect the generator from the battery when reverse current reaches the values indicated in figure 2 (see pg. 19).

3.4.1.3 Reverse voltage (not applicable to assemblies containing semiconductor devices). When tested as specified in 4.6.1.1.3, the relay assembly shall not connect the generator to the battery when the generator polarity is reversed, and subsequently shall evidence no damage resulting from the test.

3.4.1.4 Parallel operation. When tested as specified in 4.6.1.3, relay assemblies shall open and close the parelleling circuit as necessary to accomplish parallel operation. The relay assembly shall include a device to automatically open the paralleling circuit when the generator is operated as a cranking motor (i.e., when battery voltage is impressed through terminal F on the battery side of the control box).

3.4.1.5 Endurance When tested as specified in 4.6.1.1.4, the relay assembly shall evidence no burning, breakage, or other damage.

3.4.2 Voltage regulator performance (units 1 and 3 only).

3.4.2.1 Voltage adjustment range. When tested as specified in 4.6.1.2.1, the generated output voltage shall be manually adjustable within a range of 25 to 30 volts by means of a 0 to 100 ohm external rheostat (not a part of the unit).

3.4.2.2 Voltage regulation. When tested as specified in 4.6.1.2.2, the voltage regulator shall maintain output between 27.5 and 29 volts under all specified operating conditions except the initial 15-minute warmup period, and



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at speeds below minimum speed for regulation, during which the output shall be maintained between 26 and 29.5 volts. Prior to the test the voltage shall be adjusted to 28.3 ± 0.1 volts, after having reached stabilized temperature at average rated speed and no load. Voltage adjustment shall be accomplished by means of an external rheostat. With the external rheostat disconnected, output voltage shall not exceed 31.5 volts under any specified operating condition. After each change in operation, voltage shall recover to 28.3 ± 10 percent within 0.1 second and shall not develop sustained oscillations.

3.4.2.3 Resistance range. When tested as specified in 4.6.1.2.3, the voltage regulator shall evidence no instability during operation. Regulator resistance shall not exceed 1.25 ohms in the low resistance range and shall be at least 35 ohms in the high resistance range.

3.4.2.4 Parallel operation. When tested as specified in 4.6.1.3, the voltage regulator shall regulate the generator voltage such that the system output voltage shall be maintained between 26 and 29 volts. The generator employed (see 4.5.1) shall have an interpole voltage drop of from 1 to 2 volts at rated current. For regulators using current limit to control generators in parallel the system shall be stable when the load is sufficient to reduce voltage to 26 volts at rated output plus 20 percent, minus 0 percent.

3.4.2.5 Equalizer coil. When tested as specified in 4.6.1.2.4, the equalizer coil shall exhibit 0.4 ± 0.1 ohms of resistance and an amplification factor of 10 ± 2. Static type regulators shall employ a 0.4 ± 0.1 ohm resistor in place of the equalizer coil to provide compatible operation with carbon pile type regulators. In static type regulators, an amplification factor is not applicable. The current control device in the static type regulators shall limit the generator current to rated current plus 20 percent, minus 0 percent by reducing the voltage to not more than 14 volts as load resistance is reduced.

3.4.2.6 Temperature rise. When tested as specified in 4.6.1.2.5, the temperature rise of the voltage regulator shall not exceed 70°F above the temperature of the ambient still air.

3.4.2.7 Endurance. The voltage regulator shall have a minimum effective operating life of 1,000 hours with no maintenance except that one replacement of carbon disks shall be allowed if a carbon pile element is used. If semiconductor devices are used in the regulator or relay assembly, the semiconductors shall comply with Specification MIL-S-19500. When tested as specified in 4.6.1.2.6, the voltage regulator shall meet the requirements of 3.4.2.2, and regulated voltage shall drift no more than 0.1 volt during any 24-hour period nor more than 0.2 volt during the entire test for a given generator speed.

3.4.3 Waterproofness (unit 1 only). The regulator assembly shall be waterproof when tested as specified in 4.6.1.3.



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3.4.4 Vibration resistance. When tested as specified in 4.6.1.4, the units shall evidence no breakage, shorting, loosening of parts, abrasion, chattering of contacts, burning of contacts, or other damage.

3.4.5 Shock resistance. When tested as specified in 4.6.1.5, the units shall evidence no breakage, shorting, loosened or distorted parts, or other damage.

3.4.6 Extreme temperature effect. When tested as specified in 4.6.1.12, and while operating in ambient air temperatures of minus 65° ± 5°F and plus 160° ± 5°F, the unit shall meet the requirements of 3.4.1.1 through 3.4.1.3, 3.4.2.2, and 3.4.2.6, as applicable.

3.4.6.1 Extreme temperature storage. After testing as specified in 4.6.1.11, the units shall meet the requirements of 3.4.6.

3.4.7 Electromagnetic interference. The regulator assembly shall meet the requirements of MIL-STD-461 for interference control and radio suppression for tactical vehicles. Subsequent to first article approval, radio suppression acceptability shall be determined by examination to determine that suppression elements are provided in the same location and degree as in the qualified sample (see 4.3.2.2).

3.4.8 Dielectric strength (not applicable to unit 1 assemblies containing semi-conductor devices). When tested as specified in 4.6.1.7, the insulation of the unit tested shall evidence no arcing, sparking, flashover, rupture, burning, charring, smoking, or other damage.

3.4.9 Current leakage (not applicable to unit 1 assemblies containing semi-conductor devices). When tested as specified in 4.6.1.8, insulation resistance from conductor to ground, or from one conductor to another, shall be not less than 10,000 ohms after being subjected to 97 ± 3 percent relative humidity for 24 hours.

3.4.10 Fungus resistance (unit 1 only). When tested as specified in 4.6.1.10, the regulator assembly shall evidence no damage affecting mechanical or electrical operation.

3.4.11 Corrosion resistance (unit 1 only). When tested as specified in 4.6.1.13, the regulator assembly shall evidence no damage affecting mechanical or electrical operation.

3.5 Protective finish.

3.5.1 Electroplating. Except as otherwise specified, all unprotected metallic surfaces, except those of corrosion-resistant metals or alloys, shall be cadmium plated in accordance with class A of Specification QQ-P-416. Zinc



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plating, if specified, shall conform to class 1 of Specification QQ-Z-325. Silver plating, if specified, shall conform to Specification QQ-S-365, with the thickness of the silver deposited as specified.

3.5.2 Painting. Except as otherwise specified, the regulator case, cover, and base shall be prepared and painted in accordance with Drawing numbers D8689199, D8689197, and F8725895 respectively, as applicable.

3.5.2.1 Film thickness. Prime and finish coats shall have a minimum film thickness of 1 mil.

3.5.2.2 Unpainted surfaces. Screw threads shall not be painted and paint shall be omitted or removed from surfaces where required for radio-interference suppression (see 3.4.7), or for grounding of components.

3.6 Marking. Marking shall be in accordance with Standard MIL-STD-130 and the following:

3.6.1 Nameplate. Each unit 1 assembly shall be provided with a nameplate conforming to military part number B8724048. The following information shall be stamped on each nameplate.

(a) Regulator assembly, generator
(b) 24-volt, 100 to 400 ampere
(c) MS part number .....
(d) Name or trademark of manufacturer
(e) Manufacturer's serial number .....
(f) Manufacturer's part number .....
(g) Contract number .....

3.6.2 Identification marking. For identification, each relay assembly and voltage regulator assembly shall be marked with the MS part number (see 6.1) and the name or trademark of the manufacturer, as indicated on the applicable drawings or as specified by the procuring activity.

3.7 Literature. The approved wiring diagram shall be fastened on the inside of the regulator assembly cover. When specified (see 6.1), illustrated installation and maintenance, shall be supplied and packaged with each regulator assembly.

3.8 Workmanship. Workmanship on all units shall be of a quality which will assure a product free of burrs, scratches, rust, chips, and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary



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to assure supplies and services conform to prescribed requirements.

4.2 First Article Testing. The First Article sample consisting of four (4) of the units proposed to be furnished shall be submitted for testing in order to conform to 3.1. First Article testing shall be conducted by the contractor under government surveillance and shall consist of examination and classification of defects specified in 4.3.2.2 and testing specified in Table I in the order listed therein.

4.2.1 Failure. Failure of a First Article sample to pass any of the tests specified herein may be cause for the Government to refuse to witness additional testing until the faults revealed by the tests have been corrected.

4.3 Acceptance inspection.

4.3.1 Sampling for units.

4.3.1.1 Lot formation. Unless otherwise specified (see 6.1), a lot shall consist of all units of one part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for acceptance examination. Samples for acceptance examination shall be selected in accordance with Standard MIL-STD-105.

4.3.1.3 Sampling for acceptance testing. Samples for acceptance testing shall be selected in accordance with inspection level S-3 of Standard MIL-STD-105.

4.3.2 Examination inspection for units.

4.3.2.1 Acceptable quality levels. Each sample selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

| Classification | AQL |
|---|---|
| Major | 1.0 |
| Minor | 2.5 |



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Table I. Sequence of First Article Tests

| Sample No. | Paragraph No. | Test |
|---|---|---|
| | | Unit 1 - Regulator Assembly |
| 1 ..... | 4.6.1.9 | Waterproofness |
| | 4.6.1.10 | Fungus resistance |
| 2 ..... | 4.6.1.1 | Parallel operation |
| | 4.6.1.7 | Electromagnetic interference |
| | 4.6.1.11 | Extreme temperature storage |
| | 4.6.1.12 | Temperature effects |
| 3 ..... | 4.6.1.4 | Vibration |
| | 4.6.1.5 | Shock |
| | 4.6.1.6 | Current leakage |
| | 4.6.1.13 | Corrosion resistance |
| | 4.6.1.8 | Dielectric strength |
| 4 ..... | 4.6.1.2.1 | Closing voltage |
| | 4.6.1.2.2 | Opening current |
| | 4.6.1.2.3 | Reverse voltage |
| | 4.6.1.2.1 | Voltage adjustment |
| | 4.6.1.2.2 | Voltage regulation |
| | 4.6.1.2.3 | Resistance range |
| | 4.6.1.2.4 | Equalizer coil |
| | 4.6.1.2.5 | Temperature rise |
| | 4.6.1.2.4 | Endurance |
| | 4.6.1.2.2 | Voltage regulation (condition b only) |
| | | Unit 2 - Relay Assembly |
| 1 ..... | 4.6.1.2.1 | Closing voltage |
| | 4.6.1.2.2 | Opening current |
| | 4.6.1.2.3 | Reverse voltage |
| 2 ..... | 4.6.1.2.4 | Endurance |
| | 4.6.1.2.2 | Voltage regulation (condition b only) |
| 3 and 4 ..... | | Subject to unit 1 tests for samples 3 and 4 except that none of the tests given to samples 1 and 2 (above) shall be repeated. |
| | | Unit 3 - Voltage Regulator Assembly |
| 1 ..... | 4.6.1.2.1 | Voltage adjustment |
| | 4.6.1.2.2 | Voltage regulation |
| | 4.6.1.2.3 | Resistance range |
| | 4.6.1.2.4 | Equalizer coil |
| | 4.6.1.2.5 | Temperature rise |
| 2 ..... | 4.6.1.2.6 | Endurance |
| 3 and 4 ..... | | Subject to unit 1 tests for samples 3 and 4 except that none of the tests given to samples 1 and 2 (above) shall be repeated. |



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4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as follows:

| Categories | Defects | Method of inspection |
|---|---|---|
| Critical ..... | None defined. | |
| Major ..... | AQL 1.0 percent: | |
| 101 ..... | Dimension affecting interchangeability not within tolerances (see 3.3.2) | Gage |
| 102 ..... | Inadequate radio-interference suppression (see 3.4.7) | Visual |
| Minor ..... | AQL 2.5 percent: | |
| 201 ..... | Dimensions not affecting interchangeability not within tolerance (see 3.3.2) | Gage |
| 202 ..... | Improper marking (see 3.6) | Visual |
| 203 ..... | Faulty workmanship (see 3.8) | Visual |

4.3.3 Testing inspection for units.

4.3.3.1 Acceptance tests. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in table II, as applicable, using an AQL of 6.5 on the basis of percent defective.

Table II. Acceptance tests

| Sample No. | Paragraph No. | Test |
|---|---|---|
| | | Unit 1 - Regulator Assembly (includes units 2 and 3) |
| 1 ..... | 4.6.1.2.1 | Voltage adjustment |
| | 4.6.1.1.1 | Closing voltage |
| | 4.6.1.1.2 | Opening current |
| | 4.6.1.1.3 | Reverse voltage |
| | 4.6.1.2.2 | Voltage regulation |
| | | Unit 2 - Relay Assembly |
| 2 ..... | 4.6.1.1.1 | Closing voltage |
| | 4.6.1.1.2 | Opening current |
| | 4.6.1.1.3 | Reverse voltage |
| | 4.6.1.2.2 | Voltage regulation (condition b only) |
| | | Unit 3 - Voltage regulator Assembly |
| 2 ..... | 4.6.1.2.1 | Voltage adjustment |
| | 4.6.1.2.3 | Resistance range |
| | 4.6.1.2.2 | Voltage regulation |



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4.4 Control tests.

4.4.1 Sampling for control tests. When specified (see 6.1), control test samples shall be selected at the rate of 3 of every 1000 units produced except that not less than 3 nor more than 6 units shall be selected in any 30-day period.

4.4.2 Unit control testing. Units selected in accordance with 4.4.1 shall be examined for the defects specified in 4.3.2.2, and subsequently tested as specified in table III, as applicable.

4.4.3 Failure. Failure of a control test sample to pass any specified test may be cause for the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the test have been corrected.

Table III. Control tests

| Sample | Paragraph No. | Test |
|---|---|---|
| | | Unit 1 - Regulator Assembly |
| 1 ..... | 4.6.1.7 | Dielectric strength |
| 1 ..... | 4.6.1.9 | Waterproofness |
| 1 ..... | 4.6.1.4 | Vibration |
| 1 ..... | 4.6.1.2.2 | Voltage regulation (condition b only) |
| | | Unit 2 - Relay Assembly |
| 2 ..... | 4.6.1.7 | Dielectric strength (also included are any other tests deemed necessary by the Government inspector) |
| | | Unit 3 - Voltage regulator assembly |
| 2 ..... | 4.6.1.7 | Dielectric strength |
| 2 ..... | 4.6.1.9 | Waterproofness |
| 2 ..... | 4.6.1.2.2 | Voltage regulation |

4.5 Test procedures. Unless otherwise specified herein, the test procedures listed below are applicable to all units classified.

4.5.1 Test conditions and equipment. When either unit 2 or unit 3 is specified, all tests which require operation of the unit shall be conducted with the unit installed as a component of unit 1. Unless otherwise specified herein, tests shall be conducted after regulator is stabilized at generator minimum rated speed and no load except battery, and the external rheostat adjusted to obtain a regulated voltage of 28.3 $\pm$ 0.1 volts at generator average speed and no load except battery. Generators and batteries used in connection with tests shall conform to the applicable type, grade, and class of Specification MIL-G-12604 and MIL-B-11188, respectively. Unless otherwise specified herein, voltage measurements shall be taken



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between the positive terminal of the external, manual, voltage-adjusting rheostat, or the lead intended to attach thereto, and the regulator assembly ground screw. Instruments, such as voltmeters, ammeters, and ohmmeters, shall be of the laboratory type and shall be accurate to within 0.5 percent of full scale deflection and within 1 percent of indicated values. Unless otherwise specified, tests shall be performed in ambient, still air at a temperature of 77° ± 15°F. under standard atmospheric conditions. These conditions shall be based on or corrected for standard or sea level atmospheric conditions of 29.92 inches of mercury barometric pressure, 0.39-inch of mercury water vapor pressure, and 60°F. ambient air temperature. The ambient air conditions prevailing during the test shall be recorded in the test data. Unless otherwise specified, in all tests, the battery shall be connected in the circuit.

4.5.2 Definitions (see 6.3).

4.6 Tests. Unless otherwise specified herein, the tests listed below are applicable to all units classified.

4.6.1 Regulator assembly tests.

4.6.1.1 Relay assembly tests(units 1 and 2 only).

4.6.1.1.1 Closing voltage test. The relay assembly shall be assembled in the regulator assembly, electrically connected as in intended operation and voltmeters installed across the line-contactor relay, between the line-contactor relay and the generator, and between the line-contactor relay and the battery. The generator shall be operated such that the rising generator output voltage conforms to the values shown in figure 1. Voltmeter readings shall be recorded at each 0.5 volt rise in generator output voltage and compared to figure 1 to determine conformance to 3.4.1.1. The test shall be conducted twice under the conditions specified in 4.5.1, except that the first test shall be conducted with the generator and the regulator stabilized at room temperature prior to the test, and the second test with the generator and the regulator stabilized at operating temperature.

4.6.1.1.2 Opening current test. The relay assembly shall be assembled in the regulator assembly, electrically connected as in intended operation, and an ammeter installed across the line contactor relay. The test shall be conducted under the conditions specified in 4.5.1, and the maximum reverse current indicated on the ammeter before it drops to zero shall be recorded The readings thus obtained shall be compared with figure 2 to determine conformance to 3.4.1.2.

4.6.1.1.3 Reverse voltage tests (not applicable to unit 1 assemblies containing semiconductor devices). The relay assembly shall be installed in the regulator assembly and electrically connected as in intended operation. A voltmeter shall be installed across the generator and a 5-ampere fuse installed in series



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between the line contactor relay and the battery. The system shall be operated as specified in 4.5.1 until the generator and the regulator assembly reach a stabilized operating temperature. To determine conformance to 3.4.1.3, the generator polarity shall be reversed and the generator voltage raised to 30 volts and held at that value for no less than 5 minutes. At all times, when polarity is reversed, the fuse shall remain intact indicating that the generator has not been connected to the battery.

4.6.1.1.4 Endurance test. To determine conformance to 3.4.1.5, the relay assembly shall be subjected to 50,000 cycles of operation. Each cycle shall consist of the generator field circuit being opened for 5 seconds and closed for 5 seconds by means of an automatically operated contactor in series with the field. A variable resistance shall be placed across this contactor and adjusted to such value that the field current with the contactor open will be 10 percent of the field current with the contactor closed. The generator shall be operated as specified in 4.5.1 at the minimum speed for regulation plus 75 rpm, minus zero rpm.

4.6.1.2 Voltage regulator tests (units 1 and 3 only).

4.6.1.2.1 Voltage adjustment test. The rheostat specified in 3.4.2.1 shall be connected to the unit in accordance with applicable drawing, and the unit shall be operated under the conditions specified in 4.5.1 until the regulator assembly and the generator reach a stabilized operating temperature. While operating under these conditions and at a generator speed 500 rpm above the minimum speed for regulation, the rheostat shall be adjusted to maximum and subsequently to minimum resistance and the output voltage measured in each case to determine conformance to 3.4.2.1.

4.6.1.2.2 Voltage regulation test. The voltage regulator shall be assembled in the regulator assembly, electrically connected as in intended operation, and a voltmeter installed across the generator. The system shall be operated for 15 minutes to achieve a stabilized operating temperature and voltage recorded at 3-minute intervals. To determine conformance to 3.4.2.2, the voltage regulator shall be tested under the conditions specified in 4.5.1 and as follows:

(a) At minimum speed for regulation and at loads ranging from zero to full rated load including loads of 10 and 50 percent, loads shall be suddenly applied and suddenly released (shock loads) and voltage recorded immediately after application and immediately after release of load.
(b) At a speed midway between minimum speed for regulation and generator maximum rated speed, under the conditions specified in (a).
(c) At generator maximum rated speed, under the conditions specified in (a).



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(d) At no load, with generator speed varied three times between 80 percent of minimum speed for regulation and generator maximum rated speed.
(e) At 50 percent load, through the speed range specified in (d).
(f) At full rated load, through the speed range specified in (d).
(g) Under all conditions specified in (a) through (f) with the external rheostat disconnected.
(h) Carbon pile regulators intended for use with 300 ampere DC generators shall meet voltage regulation requirements with generator speed range of 2500 RPM to 8000 RPM. Voltage measurements shall be taken as specified in 4.5.

4.6.1.2.3 Resistance range test. The voltage regulator shall be assembled in the regulator assembly, electrically connected as in intended operation, and a voltmeter and an ammeter installed to indicate generator output. To determine conformance to 3.4.2.3, the voltage regulator shall be tested under the conditions specified in 4.5.1 and as follows:

(a) Following the test specified in 4.6.1.2.2, the generator shall be operated at 125 percent of minimum speed for regulation (see 4.5.2) while delivering rated current. After the generator and the regulator assembly have achieved stabilized operating temperature (see 4.5.2), the load circuit shall be opened and closed 5 times. A reading of generator voltage at rated current shall be recorded. While maintaining rated current, the generator speed shall be gradually decreased. The load circuit shall be opened and closed during this process until the generator voltage, at rated current, shall have decreased 1.5 volts below the value previously observed. At this point, the current through and the voltage drop across the voltage regulator shall be measured and the low value of resistance range calculated.
(b) With the generator driven at maximum rated speed under no load, the voltage shall be reduced by means of the external, manually adjusted rheostat, while the resistance of the voltage regulator is repeatedly determined, until the high value specified in 3.4.2.3 is attained.
(c) If arcing, output fluctuation, or instability of regulating resistance is noted, the regulator shall be operated for 50 hours and then tested as specified in 4.6.1.2.2 to determine conformance to 3.4.2.2.



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4.6.1.2.4 Equalizer coil test. To determine conformance to 3.4.2.5, the resistance of the equalizer coil shall be measured. Subsequently, the generator shall be operated at the minimum speed for regulation until the generator and the regulator assembly reach a stabilized operating temperature. While in operation, a potential of 0.15 volts shall be applied to the equalizer coil terminals of the voltage regulator in such manner as to make the positive poles coincident and the increase in generator voltage recorded. The same 0.15 volt potential shall subsequently be applied again except that the polarity shall be reverse and the decrease in generator voltage recorded Since static type regulators employ current limiting instead of a paralleling circuit, unit 1 assemblies containing static type regulators shall be tested in the following manner:
The generator shall be operated at mid-range speed. Load resistance shall be reduced until the voltage drop at the battery terminals has dropped to 14 ± 1.0 volts. The maximum and minimum load currents shall be recorded between 25 and 14 volts. Maximum generator current shall be 100 pe cent, plus 20 percent minus 0 percent of the generator rated output.

4.6.1.2.5 Temperature rise test. To determine conformance to 3.4.2.6, a thermocouple shall be inserted through an unused connector socket, or through an opening drilled th ough the drain plug for the thermocouple leads. The inside air temperature shall be measured during testing, and after stabilization, as specified in 4.6.1.2.4. Operation shall be at rated voltage and rated load until stabilized.

4.6.1.2.6 Endurance test. To determine conformance to 3.4.2.7, the voltage regulator shall be operated for 1000 hours at no load (except battery) for a minimum of 25,000 cycles. During each cycle the maximum rated generator speed shall be reached. The lowest speed eached shall be low enough to cause the contactor to open. Cycling shall be a ranged in such manner that neither operating time above nor operating time below middle speed amounts to less than 40 percent of total operating time.

4.6.1.3 Parallel operation. To determine conformance to 3.4.1.4 and 3.4.2.4, regulators shall be installed as components of two complete generating systems and connected for parallel operation. One of the generators used shall have a starting winding, and this generator shall be caused to operate as a cranking motor at some time during the test. The output voltages of the units shall be adjusted by means of an external heostat such that the value for one unit is 1 volt less, and the value of the other unit is 1 volt greater than the value specified in 4.5.1. The units shall subsequently be operated under the speed and load conditions specified in 4.6.1.2.2, except that the generator with the starting winding shall run at rated speed.

4.6.1.4 Vibration test. To determine conformance to 3.4.4, the unit shall be electrically connected to the generator and batteries as in intended operation. The unit shall be mounted on an approved vibration machine, incorporating shock mounts, if furnished, and shall be vibrated by simple harmonic motion in the direction of each of the three major axes through each of the following frequency ranges, and travels, for the length of time specified:



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| Frequency range, cps (cycles per second) | Total travel (inches, max.) | | Time (hours) |
|---|---|---|---|
| | Nonoperating | Operating | |
| 10-20 ..... | 0.100 | 0.025 | 2 |
| 20-30 ..... | 0.050 | 0.015 | 2 |
| 20-55 ..... | 0.015 | 0.003 | 2 |

The vibration machine shall be adjusted to yield the indicated maximum travel at some point within the specified frequency range. For each length of travel indicated, the entire frequency range shall be traversed each minute. During the first 1 1/2 hours of each 2-hour period, the unit shall not be operated. During the last half hour, the generator shall be operated at maximum speed under the conditions specified in 4.5.1. During and after the test the unit shall be examined for the defects specified in 3.4.4.

4.6.1.5 Shock test. To determine conformance to 3.4.5, the regulator assembly shall be tested in accordance with method 205B of Standard MIL-STD-202 using test condition C. If shock mounts are furnished, they shall be incorporated in the mounting for test.

4.6.1.6 Electromagnetic interference test. To determine conformance to 3.4.7, the regulator assembly shall be tested in accordance with the applicable provisions of MIL-STD-461A. Subsequent to testing acceptability shall be determined by examination as specified in 3.4.7.

4.6.1.7 Dielectric strength test (not applicable to unit 1 assemblies containing semiconductor devices). To determine conformance to 3.4.8, the unit classified shall be opened and subjected to conductor-to-conductor, and conductor-to-ground tests of 500 volts rms at a frequency of 60 cycles per second (cps) applied for 1 minute.

4.6.1.8 Current leakage test (not applicable to unit 1 assemblies containing semi-conductor devices). To determine conformance to 3.4.9, conducting cables, connectors, insulated terminals, windings and other conducting parts shall be subjected to appropriate 500-volt bridge o megohmmete tests.

4.6.1.9 Waterproofness test. To determine conformance to 3.4.3, the regulator assembly shall be subjected to 6 pounds per square inch (psi) internal pressure for 30 minutes and subsequently to 6 psi internal vacuum for 30 minutes, while totally submerged in a 5-percent salt-water solution. After submersion, the regulator assembly shall be subsequently tested as specified in 4.6.1.2.2 for conformance to 3.4.2.2.

4.6.1.10 Fungus resistance test. To determine conformance to 3.4.10, the regulator assembly shall be subjected to the fungus resistance test specified in Specification MIL-F-13927, except that the duration of the test shall be a single, uninterrupted period of 90 days.



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4.6.1.11 Extreme temperature storage. To determine conformance to 3.4.6.1, the unit classified shall be stored at a temperature of minus 80° ± 5°F. for a period of 72 hours and subsequently tested as specified in 4.6.1.12.

4.6.1.12 Temperature effects test. To determine conformance to 3.4.6, the unit classified shall be tested as specified in 4.6.1.1.1 through 4.6.1.1.3, 4.6.1.2.2 and 4.6.1.2.5, except that ambient air temperatures during testing shall be minus 65° ± 5°F. and plus 100° ± 5°F.

4.6.1.13 Corrosion resistance test. To determine conformance to 3.4.11, the regulator assembly shall be subjected to the method 101, 20 percent salt spray test of Standard MIL-STD-202, except that the test duration shall be 200 hours.

4.7 Inspection of preparation for delivery.

4.7.1 Materials and processes. The Government inspector shall, at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery to determine conformance to the requirements of section 5 and specifications referenced therein. Any evidence of deviation from specified requirements shall be cause for refusal of the Government inspector to accept subsequent lots until objective evidence has been provided by the supplier that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be in accordance with the applicable packaging data sheets specified by the procuring activity (see 6.1).

5.2 Marking. In addition to any special marking required (see 5.1), packages and exterior containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Unit, classification (see 1.2).
(c) Cable requirements, if different (see 3.3.5).
(d) MS part number (see 3.6.1 and 3.6.2).
(e) If literature is required (see 3.7), and when required, address of approving activity.
(f) Lot formation, if different (see 4.3.1.1).
(g) Whether or not control testing is required (see 4.4.1).
(h) Applicable packaging data sheet (see 5.1).
(i) Any special marking requirements (see 5.2).



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6.2 Definitions. The meaning of terms (see 4.5.2)such as cut-in speed, minimum rated speed, minimum speed for regulation, maximum rated speed, and maximum speed for regulation, will be defined in Specification MIL-G-12604. Other definitions are as follows:

6.2.1 Stabilized. The unit shall be considered to be stabilized when its temperature changes no more than 1°F from the initial recorded value in a period of 20 minutes.

6.2.2 Average rated speed. Average rated speed is that speed midway between minimum rated speed and maximum rated speed.

Custodians:
Army - AT
Navy - MC
Air Force - 82

Preparing activity:
Army - AT

Project No 2920-0342



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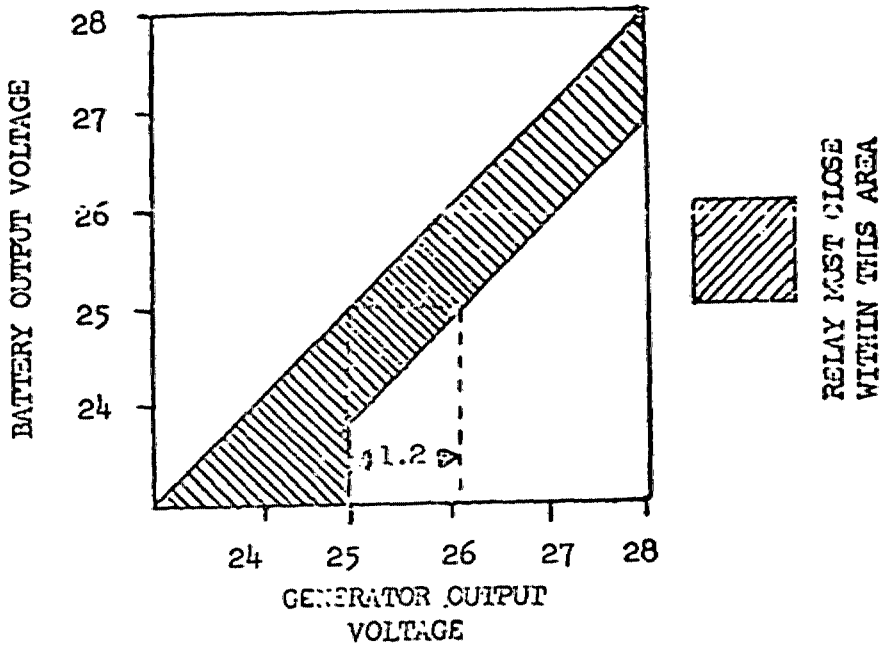



Figure 1. Relay closing voltage.

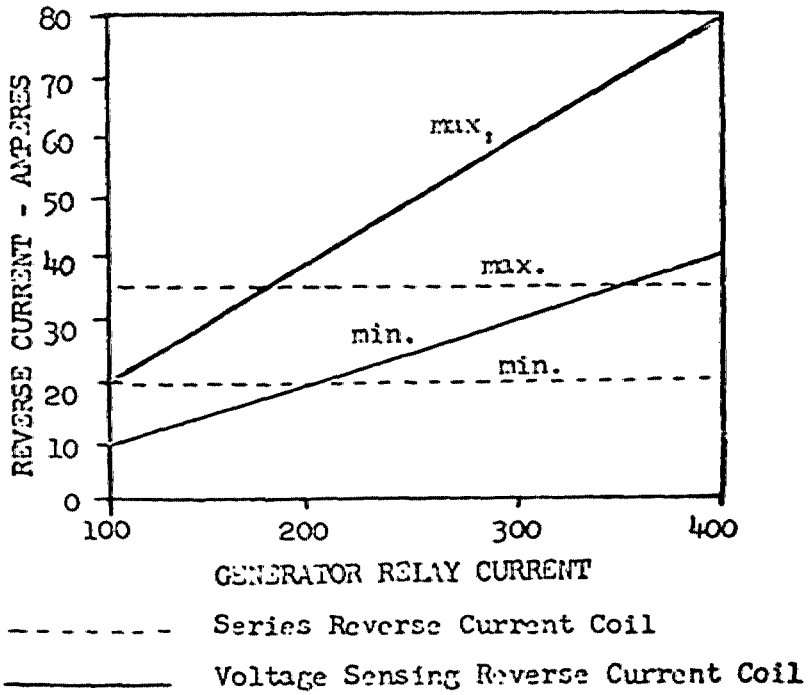



Figure 2. Relay opening current.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

DIRECT GOVERNMENT CONTRACT SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

YES NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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